

Gold Future/ETF Analysis

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Table 1: New Data: A summary of the regression coefficients and measures of goodness of fit for regressing one-day returns of 1, 2, 6 and 12-month futures on 1-day returns of spot gold.

Maturity	Slope	Intercept	R^2	RMSE
1-Month	0.98218	-9.0480×10^{-5}	0.92459	0.00269
2-Month	0.98709	-1.2547×10^{-4}	0.92244	0.00275
6-Month	0.98870	-9.0354×10^{-5}	0.91718	0.00285
12-Month	0.92794	9.3489×10^{-5}	0.76020	0.00520

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Table 2: Leung & Ward (2015) Data: Regression coefficients for one-day futures returns on one-day spot gold returns.

Maturity	Slope	Intercept	R^2	RMSE
1-Month	0.94314	2.4191×10^{-5}	0.80947	0.00530
2-Month	0.94301	1.0000×10^{-5}	0.80911	0.00530
6-Month	0.94348	3.1797×10^{-5}	0.80934	0.00530
12-Month	0.94358	-5.2333×10^{-5}	0.80984	0.00529

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Table 3: New Data: Slopes and R^2 from the regressions of futures returns versus gold returns over different holding periods.

	Days	1-Month	2-Month	6-Month	12-Month
Slope	1	0.96061	0.96608	0.96757	0.91176
	5	0.99723	1.00073	0.99971	0.94958
	10	0.99556	0.99782	0.99821	0.97219
	15	0.99821	1.00003	0.99906	0.98087
R^2	1	0.87699	0.87470	0.87272	0.73603
	5	0.96413	0.96345	0.96008	0.87631
	10	0.98020	0.97931	0.97586	0.93513
	15	0.98603	0.98532	0.98191	0.95316

Table 4: Leung & Ward (2015) Data: Slopes and R^2 from regressions of futures returns vs. gold returns over different holding periods.

	Days	1-Month	2-Month	6-Month	12-Month
Slope	1	0.94314	0.94301	0.94348	0.94358
	5	0.99805	0.99752	0.99757	0.99715
	10	1.01075	1.01020	1.00970	1.00841
	15	0.99448	0.99461	0.99431	0.99345
R^2	1	0.80947	0.80911	0.80934	0.80984
	5	0.97154	0.97158	0.97194	0.97165
	10	0.98516	0.98530	0.98572	0.98550
	15	0.98704	0.98691	0.98725	0.98713

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The out of sample period is defined as starting on January 1, 2024 and is approximately 10% of the data. The RMSEs are calculated in dollar terms assuming a notional investment of \$1000 in the spot gold price. The weights are calculated using a constrained least squares regression that forces the weights to sum to one. The GLD RMSE is \$6.00 for the out of sample period.

Table 5: New Data: Weights and in/out of sample RMSEs for portfolios of 1, 2, 3, and 4 futures contracts.

Portfolio	w_0	w_1	w_2	w_6	w_{12}	RMSE _{in}	RMSE _{out}
1-m	0.00285	0.99711				\$3.27	\$9.18
2-m	0.00723		0.99273			\$4.22	\$10.64
12-m	0.04846				0.95208	\$19.36	\$24.98
1-m, 2-m	0.00189	1.22277	-0.22470			\$3.23	\$8.91
1-m, 6-m	0.00198	1.04254		-0.04456		\$3.24	\$8.90
1-m, 12-m	0.00234	1.01049			-0.01286	\$3.26	\$9.16
2-m, 6-m	0.00367		1.22476	-0.22854		\$3.81	\$9.41
2-m, 12-m	0.00390		1.08780		-0.09172	\$3.87	\$10.60
6-m, 12-m	0.00919			1.66574	-0.67435	\$7.00	\$23.51
1-m, 2-m, 6-m	0.00184	1.20060	-0.19291	-0.00956		\$3.23	\$8.88
1-m, 2-m, 12-m	0.00204	1.28345	-0.29678		0.01126	\$3.22	\$8.84
1-m, 6-m, 12-m	0.00210	1.10380		-0.17056	0.06457	\$3.21	\$8.31
2-m, 6-m, 12-m	0.00369		1.22868	-0.23595	0.00348	\$3.81	\$9.37
1-m, 2-m, 6-m, 12-m	0.00196	1.24924	-0.17947	-0.13469	0.06287	\$3.20	\$8.30

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Table 6: Leung & Ward (2015) Data: Weights and in/out of sample RMSEs for portfolios of 1 and 2 futures contracts.

Portfolio	w_0	w_{short}	w_{long}	RMSE _{in}	RMSE _{out}
1-m	-0.01071	1.01071		\$6.63	\$3.34
2-m	-0.04835	1.04835		\$12.22	\$6.12
12-m	-0.06842	1.06842		\$15.11	\$5.71
1-m, 2-m	-0.00088	1.27315	-0.27227	\$6.27	\$2.79
1-m, 6-m	-0.00171	1.23899	-0.23729	\$6.27	\$2.98
1-m, 12-m	-0.00021	1.19336	-0.19315	\$6.29	\$3.00
2-m, 6-m	-0.04079	5.06602	-4.02523	\$10.27	\$9.37

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Table 7: New Data: Regression coefficients for one-day returns of (L)ETFs versus spot gold.

(L)ETF	Leverage (β)	Slope	Intercept	t-stat	p-value	R^2	RMSE
GLD	1	1.00061	-8.8911×10^{-6}	0.25954	0.79523	0.97140	0.00185
UGL	2	2.02088	-1.9483×10^{-4}	2.77734	0.00552	0.96438	0.00349
GLL	-2	-2.01429	1.8890×10^{-4}	-1.92313	0.05457	0.96497	0.00345
DGLD	-2	-1.97565	-8.4896×10^{-5}	0.46841	0.63954	0.40656	0.02098
UGLDF	3	3.00632	4.5874×10^{-4}	0.05822	0.95358	0.25518	0.04463

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Table 8: Leung & Ward (2015) Data: Regression coefficients for one-day returns of (L)ETFs versus spot gold.

(L)ETF	Slope	Intercept	t-stat	p-value	R^2	RMSE
GLD	1.00540	-1.6427×10^{-6}	1.42692	0.15382	0.98060	0.00163
UGL	2.00572	-1.3107×10^{-4}	0.73319	0.46357	0.97930	0.00337
GLL	-2.00556	-1.0550×10^{-4}	0.67089	0.50240	0.97673	0.00358
UGLD	2.99358	-1.9860×10^{-4}	0.45211	0.65133	0.98484	0.00421
DGLD	-2.97528	-1.6912×10^{-5}	0.97442	0.33019	0.95256	0.00752

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The t-statistics far from zero for UGL and GLL indicate that the slopes are statistically significantly different from the specified leverage ratios.

Table 9: New Data: Weights and in/out of sample RMSEs for portfolios replicating leveraged benchmarks.

Benchmark	Portfolio	w_0	w_1	w_2	w_6	w_{12}	RMSE _{in}	RMSE _{out}
2x Leveraged	1-m	-1.62098	2.62436				\$58.87	\$331.76
	2-m	-1.61086		2.61411			\$56.14	\$336.05
	6-m	-1.56497			2.56792		\$45.85	\$357.43
	12-m	-1.49557				2.49767	\$46.46	\$375.93
	1-m, 2-m	-1.56365	-10.84072	13.40753			\$50.70	\$353.93
	1-m, 6-m	-1.54357	-1.39403		3.94189		\$45.17	\$365.64
3x Leveraged	1-m	-4.03105	5.03349				\$188.10	\$1036.37
	2-m	-4.01308		5.01515			\$183.04	\$1042.61
	6-m	-3.92928			4.92945		\$161.06	\$1076.52
	12-m	-3.79391				4.79256	\$147.85	\$1116.16
	1-m, 2-m	-3.85870	-35.37062	40.23119			\$165.26	\$1085.79
	1-m, 6-m	-3.79649	-7.19724		11.99866		\$149.02	\$1111.73

Table 10: Leung & Ward (2015) Data: RMSEs for portfolios replicating leveraged benchmarks.

Benchmark	Portfolio	RMSE _{in}	RMSE _{out}
+2x (UGL)	1-m	\$153.20	\$41.49
	2-m	\$140.41	\$49.57
	6-m	\$138.63	\$47.46
	12-m	\$134.51	\$48.30
+3x (UGLD)	1-m	\$555.49	\$111.51
	2-m	\$529.91	\$125.99
	6-m	\$526.58	\$122.33
	12-m	\$518.97	\$123.83